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The OIA Reference Model — Executive Accessibility Enhancement

v0.5.1 Working Draft · Internal review · 2026

Three additions to make the OIA more directly engageable by executive audiences: a one-page executive overview, a “why it matters” line for every layer and span, and four explicit named connection callouts. Applied first to the canonical knowledge base structure; promotable into the Reference Paper and downstream materials once approved.

How to use this draft

This document holds three proposed additions to the OIA v0.5 body of work. None is architectural — the layers, spans, threats, properties, and lens roster of v0.5 are preserved exactly as published. What changes is the editorial surface around them: short consequence-forward lines, an executive-legible one-page entry point, and explicit naming of the cross-layer patterns that are currently expressed only implicitly through span concentration callouts.

The three additions are designed to be evaluated independently. Each could be promoted into the Reference Paper alone, or all three together. The recommended sequence is:

1. **The “why it matters” lines** — lowest effort, highest distribution. Once approved, they appear in the Reference Paper layer chapters, the canonical knowledge base, the Visual Reference, the executive deck, the OIA Prism app cards, and the architecture-with-frameworks map.
2. **The one-page executive overview** — drops into the front of the Reference Paper as a

new Section 1.5 (or equivalent), and into the canonical knowledge base as a new top-level section.

3. **The four named connection callouts** — appear as sidebars in the Reference Paper, callouts in the Visual Reference, and standalone slides in the executive deck.

When the draft is approved, the canonical knowledge base receives the additions first (single source of truth), then the downstream materials are reconciled to match.

Part 1 — One-page executive overview

Designed to be the page an executive reads if they read nothing else. Earns the deeper read of the Reference Paper, the Lens that fits their role, and the canonical knowledge base. Drafted to fit on a single printed page in the OIA design system.

The OIA Reference Model — In Brief

The Open Intelligence Architecture describes the structural shape of enterprise intelligent systems. Ten layers, seven cross-layer spans, three defining properties, four named threat surfaces. Vendor-neutral, audience-neutral, and additive to the frameworks the enterprise already uses.

Three defining properties of intelligent systems

- *Persistence* — systems that maintain state across interactions.
- *Autonomy* — systems that act with degrees of authority humans didn't pre-approve case-by-case.
- *Consequence* — outputs and actions whose correctness must be demonstrable against current threat reality, not just at release.

Ten layers, numbered 0 through 9

- *Layer 0 · Physical Compute* — the physical substrate: semiconductors, energy, facilities, geographic siting.
- *Layer 1 · Silicon Abstraction* — hardware diversity behind portable runtimes and trusted execution primitives.
- *Layer 2 · Sovereign Infrastructure* — compute, storage, networking with sovereignty made explicit.
- *Layer 3 · Agent Data Substrate* — the persistent, auditable, access-controlled data environment.

- *Layer 4 · Model Training and Adaptation* — pre-training, fine-tuning, RLHF, and continuous adaptation.
- *Layer 5 · Inference and Retrieval* — where models run and where their trustworthiness becomes a live operational concern.
- *Layer 6 · Context and Knowledge* — retrieval, grounding, continuous perception.
- *Layer 7 · Orchestration and Workflow* — multi-step agent action, tool invocation, the threat surface specific to the agentic era.
- *Layer 8 · Continuity Fabric* — cognitive state preserved through change. Recovery, rollback, post-incident integrity.
- *Layer 9 · Human and Browser Interface* — where humans and intelligent systems interact.

Seven cross-layer spans

- *Security* — the widest concern; threads through every layer.
- *Sovereignty* — three dimensions: jurisdictional, operational, energy.
- *Auditability* — the capacity to reconstruct, after the fact, what happened and why.
- *Identity* — three actors require continuous trust governance: humans, agents, foundation models.
- *Energy and Environmental Concerns* — architectural rather than just operational; originates at Layer 0.
- *Provenance* — lineage of information, models, decisions, actions.
- *Verifiability* — operations cryptographically proven, not contractually asserted. New in v0.5.

Four named threat surfaces of the agentic era

- *Mimicry* — telling humans from agents at every entry point.
- *Agent-in-the-middle* — adversarial intermediaries between agents and tools. The agentic SSL strip.
- *Agent spoofing* — agent identities inheriting trust they shouldn't have.
- *Model compromise* — the trusted model that is no longer trustworthy.

Composition with existing frameworks

The OIA composes with NIST CSF, NIST AI RMF, MITRE ATLAS, MITRE D3FEND, OWASP Top

10 for LLM Applications, the Confidential Computing Consortium body of work, and ISO/IEC 42001. The OIA provides the structural map within which these frameworks compose. It is additive, not competitive.

The OIA Lens System translates the architecture for the audiences accountable for AI inside the enterprise. The architecture is invariant; the lens overlay changes per audience. Three lenses are published with v0.5 — the CISO Lens, the CIO/CTO Lens, the Board and Risk Lens. Three are anticipated — CFO, General Counsel and Compliance, CHRO.

Part 2 — “Why it matters” lines

One sentence per layer and per span. Audience-neutral, stakes-forward, written to convert each architectural element from a description into a proposition the reader has to take a position on. Each line is short enough to live in body text, in the Visual Reference, in the executive deck, in the OIA Prism app cards, and in the architecture-with-frameworks map.

For the ten layers

Layer 0 · Physical Compute. *If your AI’s physical substrate is in the wrong jurisdiction, no amount of architectural sovereignty above can recover the posture.*

Layer 1 · Silicon Abstraction. *Verifiable trust starts here — without attested execution at the silicon, every claim about model integrity above is contractual rather than proven.*

Layer 2 · Sovereign Infrastructure. *Sovereignty can’t be retrofitted higher in the stack. If it isn’t addressed here, it’s gone everywhere above.*

Layer 3 · Agent Data Substrate. *This is where AI’s data exposure is either governed or isn’t — including the embedded LLM channels traditional DLP wasn’t built to see.*

Layer 4 · Model Training and Adaptation. *Training data and fine-tuning processes are where future regulatory liability is created or contained.*

Layer 5 · Inference and Retrieval. *The model that decided is no longer necessarily the model deployed. Continuous trust scoring is now an operational discipline, not a procurement check.*

Layer 6 · Context and Knowledge. *Adversaries who control the context window control the model output. Prompt injection lives here.*

Layer 7 · Orchestration and Workflow. *Three of the four agentic-era threat surfaces land here. Agent identity governance either exists or doesn’t.*

Layer 8 · Continuity Fabric. *Without it, an agent that mis-acted has no audit trail and no*

recovery path — and the enterprise’s cognitive state is owned by whichever framework or model provider got there first.

Layer 9 · Human and Browser Interface. *If you can’t tell humans from agents at every entry point, every other layer of governance can be bypassed at the door.*

For the seven cross-layer spans

Security. *AI security failures unfold at machine speed and across an attack surface no prior security architecture was designed for.*

Sovereignty. *When sovereignty is contractual rather than architectural, it surfaces only at the audit or the incident — usually too late.*

Auditability. *An auditor’s first question is “show me the evidence.” If auditability wasn’t designed in, the answer is “we don’t have it.”*

Identity. *Foundation models and agents are now actors inside the enterprise. The identity infrastructure governing humans wasn’t built to govern them.*

Energy and Environmental Concerns. *Energy is now an architectural constraint on AI strategy, not just an operating cost. Where workloads run and at what scale follows from grid capacity, sovereignty, and disclosure regimes.*

Provenance. *Provenance is what makes auditability possible. Where it’s missing at one layer, the audit chain breaks at every layer above.*

Verifiability. *When trust is contractual, it can be broken silently. When trust is verifiable, it can’t.*

Part 3 — The four named connections

Cross-layer patterns currently expressed only implicitly through span concentration callouts. Naming them explicitly gives executives the architectural patterns to engage with, rather than asking them to reconstruct the patterns from the per-layer chapters. Each callout is three or four sentences — short enough to live as a sidebar, a Visual Reference annotation, or a standalone slide.

Connection 1 — The State-Holding Pair (Layers 3 and 8)

The OIA holds enterprise cognitive state in two architecturally distinct layers, deliberately bracketed around the operational stack. Layer 3 (Agent Data Substrate) is the persistent substrate below — the place where data, embeddings, and operational records live. Layer 8 (Continuity Fabric) is the active fabric above — the place where cognitive state is preserved

through change, where recovery and rollback happen.

Why it matters. Conflating these two ways intelligent systems hold state produces architectures that fail at the worst possible moment. An agent framework that holds its own state without a substrate fails the audit. A data substrate with no continuity fabric above it leaves the enterprise without recovery when an agent mis-acts. The asymmetric pairing — substrate below, fabric above — is the architectural pattern that makes both auditable history and active recovery possible.

Reference: Canonical knowledge base §3.3; Reference Paper layer chapters for Layers 3 and 8.

Connection 2 — The Threat Concentration (Layers 5, 7, 9)

Three of the four named threat surfaces of the agentic era originate or propagate through the operational stack — not at the perimeter where prior security architectures expected attacks to land. Mimicry lands at Layer 9 and propagates downward. Agent-in-the-middle and agent spoofing concentrate at Layer 7. Model compromise originates at Layer 5.

Why it matters. The security architecture of the agentic era cannot be built at the edge. The threats live where the work happens — inside the operational stack, where humans, agents, and models interact. An enterprise that hardens the perimeter and leaves the operational stack untouched has defended against an earlier era's threats. This is the architectural argument for AI security investment being weighted toward Layers 5, 7, and 9, not the network and identity perimeter.

Reference: Canonical knowledge base §6; Visual Reference — Four Named Threat Surfaces diagram; CISO Lens §4.

Connection 3 — The Verifiability Chain (Layers 1, 2, 4, 5, 8)

Verifiable trust is anchored at Layer 1 by hardware-attested execution primitives — Intel TDX, AMD SEV-SNP, NVIDIA Confidential GPU, ARM CCA, and the Confidential Computing Consortium body of work. It operationalises at Layer 2 as attested compute, storage, and networking. It extends at Layer 4 to attested fine-tuning for regulated adaptation. It realises at Layer 5 as attested inference for high-assurance workloads. It reaches Layer 8 as witness-chain logic for verifiable continuity through change.

Why it matters. Verifiability is the property that turns trust from a contractual claim into a cryptographic proof. For regulated workloads — finance, defence, healthcare — and for sovereign AI deployments where the operator must not need to be trusted, the chain is what makes the workload viable. Without the chain anchored at silicon, every claim about model integrity above is a vendor assertion. The chain is also why Verifiability was elevated to a span in v0.5: it threads through enough layers that anchoring it at just one would have

understated its architectural reach.

Reference: Canonical knowledge base §5.7; v0.5 Architecture Lock — Refinement 1.

Connection 4 — The Three-Actor Identity Pattern (Layers 5, 7, 9)

Identity in the agentic era governs three classes of actor: humans, agents, and foundation models. Each appears at a different layer in primary form — humans most acutely at Layer 9, agents at Layer 7, foundation models at Layer 5 — but each must be governed across the layers where they interact. The CISO Lens treats the three actors as the central axis of the architecture's security reading.

Why it matters. The existing enterprise identity infrastructure was built for one class of actor: humans. Agents inherit trust they shouldn't have when they're not scoped, attested, and behaviourally baselined. Foundation models change behaviour under the hood and can no longer be governed by procurement-time evaluation alone. The architecture's contribution is naming these three actors as one governance problem at three layers, rather than three separate problems addressed by three different specialist teams.

Reference: Canonical knowledge base §5.4; CISO Lens §3 and §4; v0.5 Architecture Lock — Refinement 2.

What to do with this draft

Three review questions are worth deciding on before promoting any of this into the published body of work.

On the “why it matters” lines. Each is intentionally short and intentionally stakes-forward. Read them in order and check that none feels overstated — the discipline is that every line should be defensible to a sceptical reader and not require a footnote to justify. Lines that don't pass this test get tightened or cut. The current draft errs slightly toward direct over diplomatic; if the Foundation reading is that some land too sharply for institutional voice, the right edit is sharpening the language rather than softening the stakes.

On the executive overview. The page is designed to be read in 60–90 seconds. Test it by handing it to one or two executives outside the OIA author circle and asking what they can recall five minutes later. If the recall is reliably “ten layers, seven spans, four threats, three properties,” the page is doing its job. If the recall is hazier, the page needs further compression rather than expansion.

On the four named connections. These are the highest-leverage editorial addition in the draft, but also the most consequential — once named, they become part of the architectural vocabulary executives will use, and revising them later will be harder than getting them right

now. The four chosen are the connections I judge most architecturally consequential. There are at least two other candidates worth considering — the cross-actor identity governance pattern stretched across humans, agents, and models, and the Provenance-Auditability dependency where one can't function without the other — that could be promoted from implicit to explicit in a future pass.

Recommended decision sequence. Approve or revise the "why it matters" lines first — they have the broadest distribution and the smallest editorial surface. Approve or revise the executive overview second — it shapes how the Reference Paper opens, which influences how everything downstream is read. Approve or revise the four named connections last — they're additive editorial content and don't have to ship in the same release.

If all three additions are approved without major revision, the work to promote them into the v0.5 body of work is roughly: one editorial pass over the canonical knowledge base, one over the Reference Paper, one over the Visual Reference, one over the executive deck, and one over the OIA Prism app cards. Roughly a day's work, distributed across the body of work.